

Software Requirements Specification (SRS)

Municipal Solid Waste Route Optimizer

Standard IEEE 830-1998 Specification Format | Student SRN: PES1UG24CS453

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Municipal Solid Waste Route Optimizer System. The system ingests ultrasonic fill-level telemetry from smart trash receptacles, calculates dynamically optimized collection routes for city garbage trucks under stringent real-time constraints, and provides full lifecycle tracking for missed-pickup citizen complaints.

1.2 Scope

The Municipal Solid Waste Route Optimizer replaces static garbage pickup schedules with real-time, demand-driven dispatching. It monitors 10,000 smart trash bins equipped with ultrasonic sensors, dynamically includes only receptacles over 80% full, recalculates coordinated routes for 50 collection trucks in under 45 seconds, delivers digital manifests to truck drivers, and provides citizen complaint registration.

1.3 Definitions, Acronyms, and Abbreviations

- VRP: Vehicle Routing Problem (combinatorial NP-hard optimization).
- IoT: Internet of Things (smart ultrasonic bin sensors).
- RBAC: Role-Based Access Control.
- SLA: Service Level Agreement (sub-45s route calculation constraint).
- MQTT: Message Queuing Telemetry Transport (lightweight sensor messaging protocol).
- gRPC: High-performance, open-source universal RPC framework.

2. Overall Description

2.1 Product Perspective

The system operates as a cloud-native microservices architecture comprising an API Gateway, Sensor Data Ingestion Service, Route Optimization Service, Driver Manifest Service, Complaint Management Service, and a multi-model spatial/time-series persistence tier.

2.2 User Classes and Characteristics

1. Sanitation Supervisor: Operations manager overseeing city-wide waste collection, monitoring telemetry graphs, initiating route recalculation, and inspecting complaint resolution.
2. Truck Driver: Field operator navigating in-cab navigation terminals to follow assigned collection manifests and acknowledge pickups.
3. Citizen: Public user reporting missed garbage collections or overflow incidents via citizen portal.
4. System Administrator: IT personnel maintaining gateway configurations, security certificates, and service scaling.

2.3 Operating Environment & Constraints

The system runs on containerized Linux nodes (Kubernetes) backed by PostgreSQL with PostGIS spatial extensions, TimescaleDB time-series storage, and Redis cache. Truck terminals operate on Android/iOS via progressive web application / native mobile frameworks.

3. Specific Functional Requirements

FR-1: Ultrasonic Telemetry Ingestion: The system shall continuously ingest ultrasonic fill-level readings from up to 10,000 smart waste bins over MQTT/cellular (NB-IoT), validating message checksums, timestamps, and geolocation tags.

FR-2: 80% Fill-Level Threshold Dynamic Filtering: The system shall dynamically filter active bins, queuing for pickup only receptacles reporting a verified fill level exceeding 80% capacity, discarding sub-threshold bins from collection schedules.

FR-3: Sub-45s Fleet Route Optimization: The Route Optimization Service shall solve the multi-vehicle routing problem for 50 collection trucks across 10,000 bins in strictly under 45 seconds upon trigger, minimizing fuel consumption and travel duration.

FR-4: Real-Time In-Cab Manifest Streaming: The Driver Manifest Service shall compile turn-by-turn waypoint manifests for each truck and stream active manifests to in-cab driver applications via secure WebSockets.

FR-5: Pickup Verification & Bin Status Reset: The system shall allow drivers to confirm bin clearance in real time, immediately logging pickup completion and resetting the bin status in the spatial database.

FR-6: Citizen Missed-Pickup Complaint Management: The system shall allow citizens to log missed-pickup reports with photographic evidence and GPS coordinates, generating an auditable tracking ticket with status progression.

FR-7: Supervisor Operations Dashboard & Dispatch Override: The Web Portal shall provide supervisors with an interactive GIS map displaying live truck locations, bin fill levels (>80% highlighted in amber/red), and manual route re-dispatch controls.

4. Non-Functional Requirements

NFR-1: Performance & Computational Latency: Route recalculation for 50 trucks across 10,000 candidate bins must execute in strictly less than 45 seconds (99th percentile). Ingestion latency must not exceed 250ms per telemetry burst.

NFR-2: Scalability: The ingestion tier shall horizontally auto-scale to process over 5,000 telemetry messages per second during peak reporting intervals without packet drops.

NFR-3: Reliability & Fault Isolation: Failure in the Complaint Management Service or telemetry stream shall not impede active truck navigation or the Route Optimization Service.

NFR-4: Security & Authentication: All external API traffic shall require TLS 1.3 encryption. APIs shall enforce JWT-based RBAC with strict rate limiting to prevent denial-of-service attempts.

NFR-5: Availability: The system shall guarantee a minimum service availability of 99.9% uptime during operational sanitation shifts (05:00 - 22:00 daily).

NFR-6: Data Integrity: Spatial geometries and historical telemetry records shall adhere to ACID transaction guarantees, preventing duplicate pickup assignments.

5. Security Objectives & Traceability

- Security Objective 1 (Confidentiality & Access Control): Enforce strict boundary isolation at the API Gateway; external citizens cannot view municipal telemetry data or driver manifests.
- Security Objective 2 (Data Integrity & Authenticity): Sensor telemetry packets must include HMAC signatures to prevent spoofed fill-level reports from triggering unauthorized truck dispatches.
- Security Objective 3 (Resilience & Blast Radius Control): Database schemas are segmented per microservice with principle of least privilege, preventing lateral movement in case of service compromise.